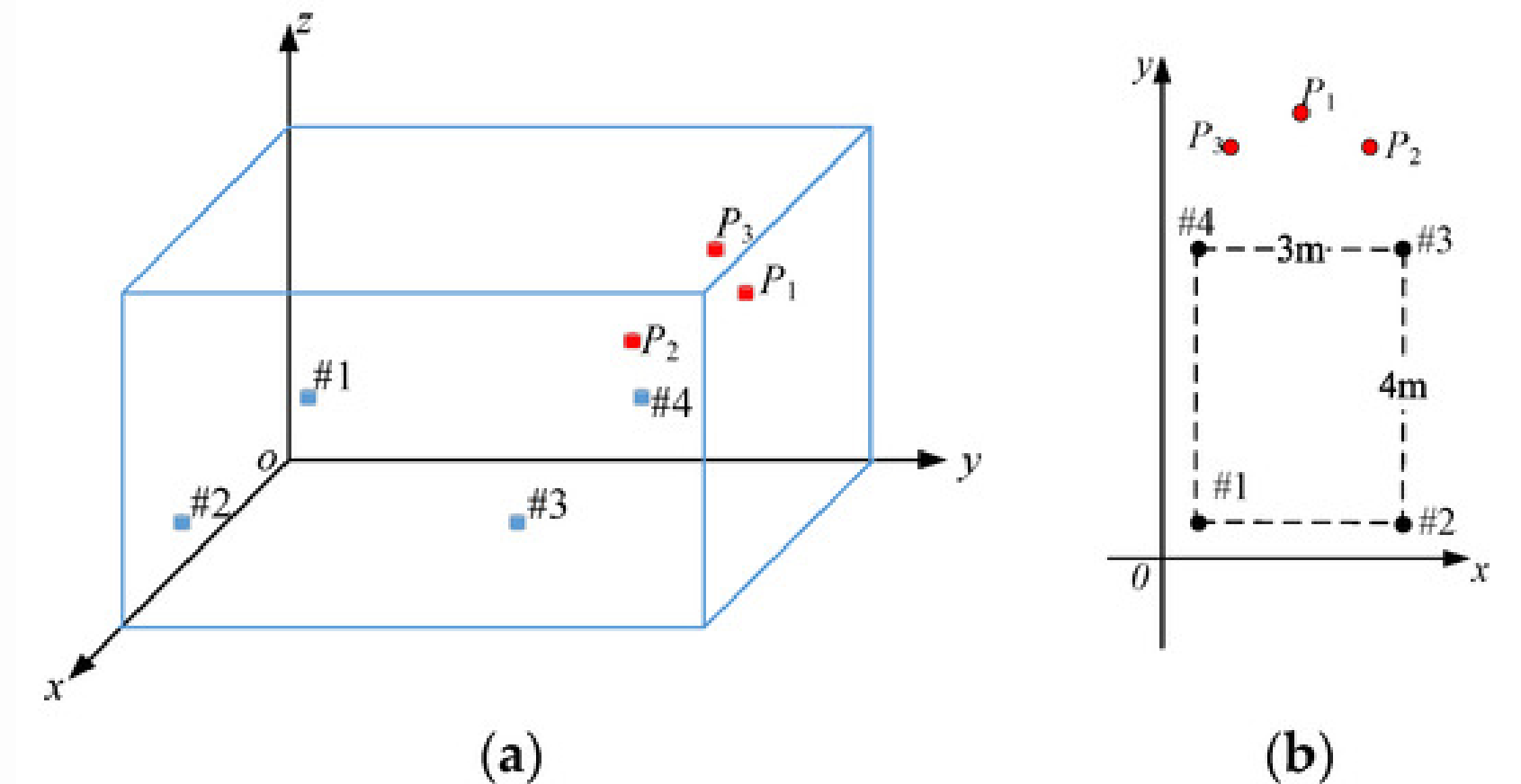


# Simultaneous Localisation of Multiple Partial Discharge Sources



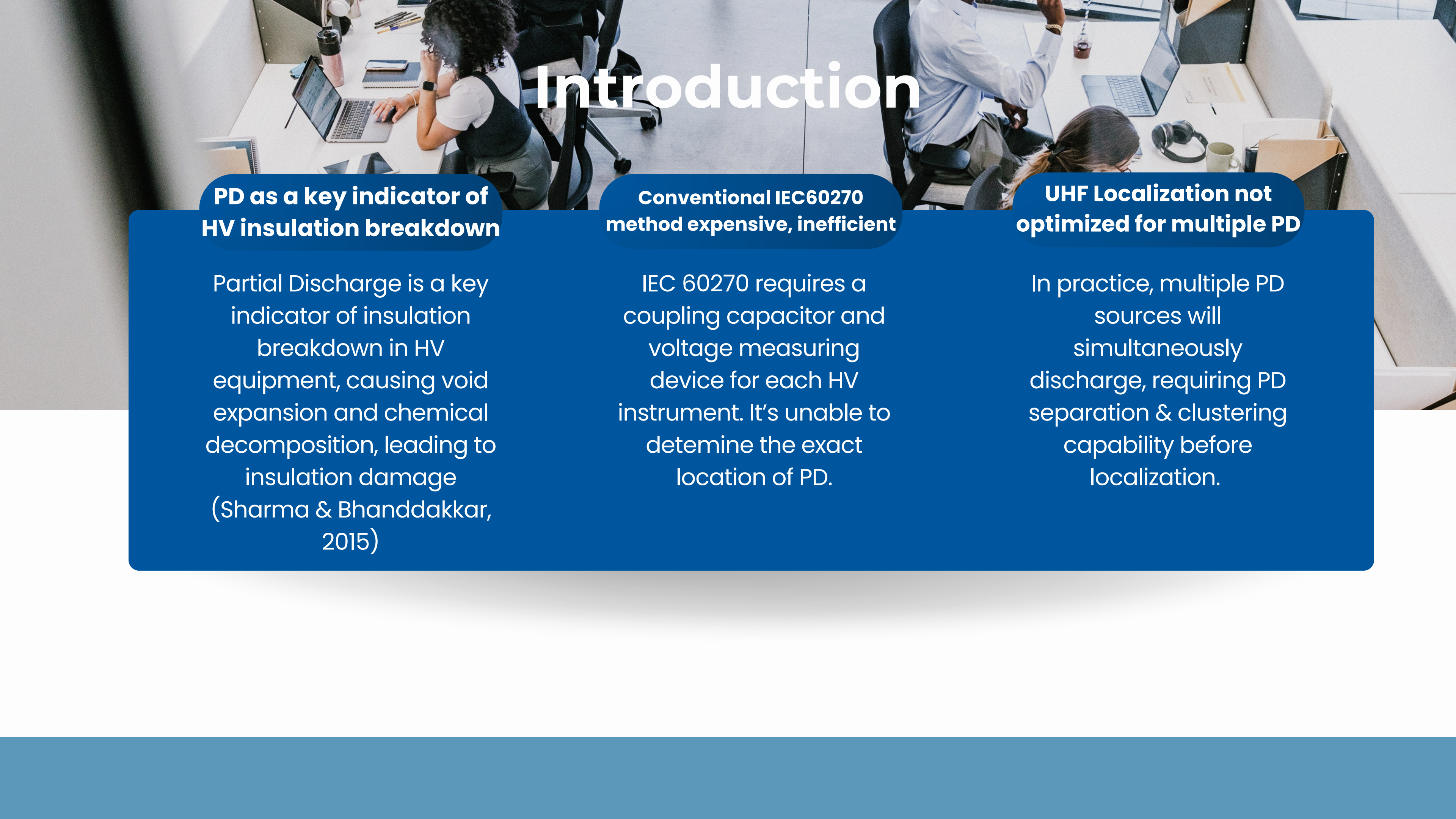
KIE4002 Final Year Project, 2025/2026, Sem 1  
Chiang Chang Zee  
Supervisor: Ir. Dr. Wong Jee Keen Raymond

Image source: He et al, (2020)

# Contents

- ▶ Intro
- ▶ Problem Statement
- ▶ Methodology
- ▶ Results & Discussions
- ▶ Future Work
- ▶ Conclusion





# Introduction

## PD as a key indicator of HV insulation breakdown

Partial Discharge is a key indicator of insulation breakdown in HV equipment, causing void expansion and chemical decomposition, leading to insulation damage (Sharma & Bhanddakkar, 2015)

## Conventional IEC60270 method expensive, inefficient

IEC 60270 requires a coupling capacitor and voltage measuring device for each HV instrument. It's unable to determine the exact location of PD.

## UHF Localization not optimized for multiple PD

In practice, multiple PD sources will simultaneously discharge, requiring PD separation & clustering capability before localization.



# The Appeal of UHF Localization

Four sensors, wide coverage, greater accuracy

Unlike traditional IEC 60270 methods that require 1 measuring set up for 1 HV equipment, UHF sensors can detect signals from a wide coverage area.

In addition, the exact location of PD can be determined, unlike IEC

## Problem: Separation of Signals

The use of one set of UHF sensors to localize many PD sources lead to the need to perform clustering. Signals can overlap and attenuate, making it difficult to tell which signals across each channel belong to the same PD

## Research Gap

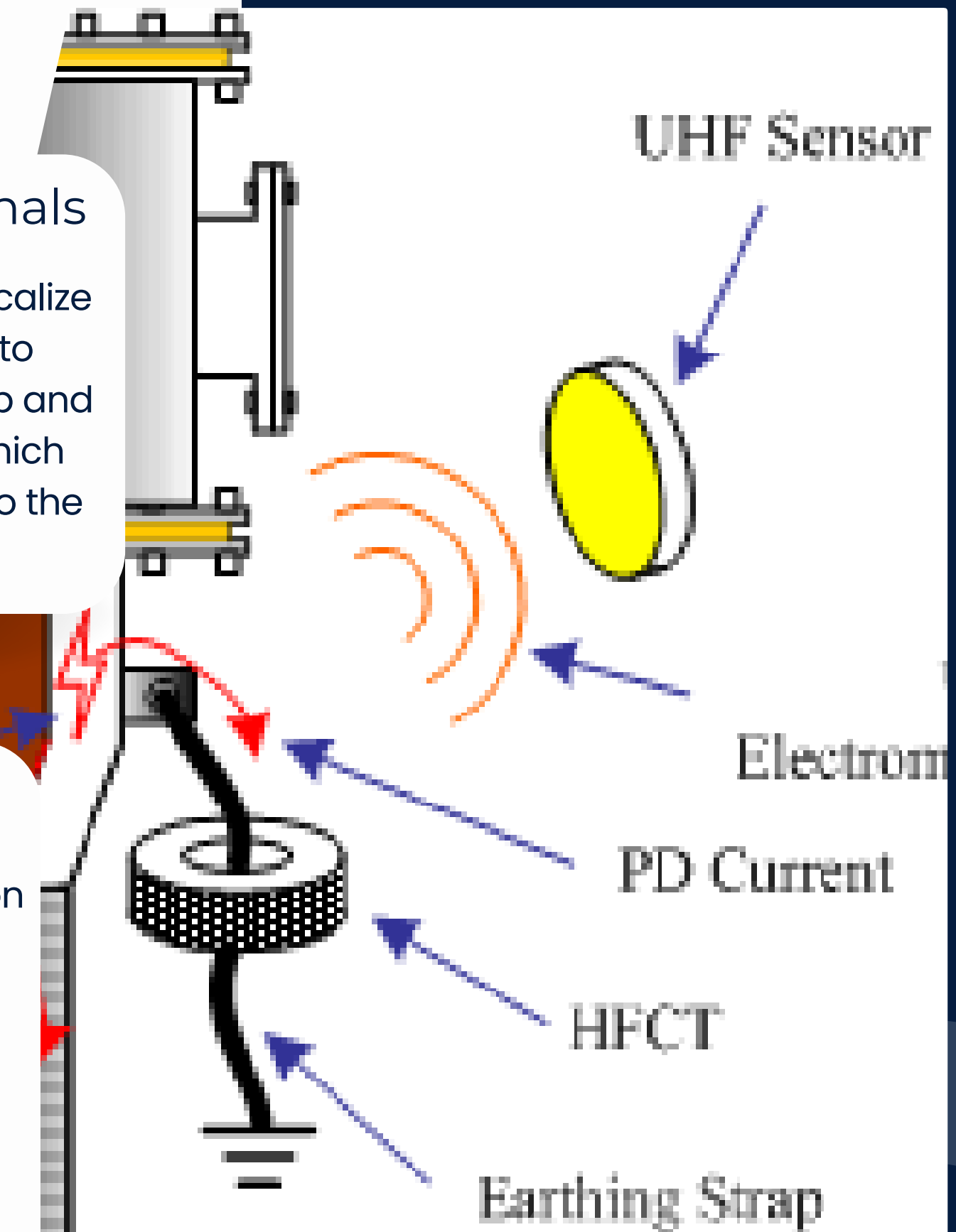
Accurate Digital Twin of PD

Current research in the area of simulating PD signals are not directly applicable to PD separation/clustering. Many are focused on denoising, propagation characteristics and the comparison of simulated and measured waveforms, which **do not place high importance on the waveform accuracy**.

Obtaining measured results for PD consist of many challenges: approval, equipment, cost.

## Underdeveloped Multiple PD UHF Localization

The research done on multiple PD localization is still in the early stages of development. Research is currently focused on many separate things: denoising, localization, separation/clustering, propagation characteristics. Has not progressed to research integrated, end-to-end solutions. In addition, current solutions are computationally expensive, requiring a lot of features and processing time.



Localization mechanism using HF



# Objectives

## Digital Twin

Develop a digital twin environment of EM wave propagation with multiple PD, incorporating white Gaussian noise. Validate by showing that there is a <20% difference in clustering accuracy between real and simulated datasets

## Clustering

Evaluate and compare the performance of different feature extraction methods:

1. Wavelet Analysis+PCA
2. Envelope analysis
3. Statistical analysis

and clustering algorithms:

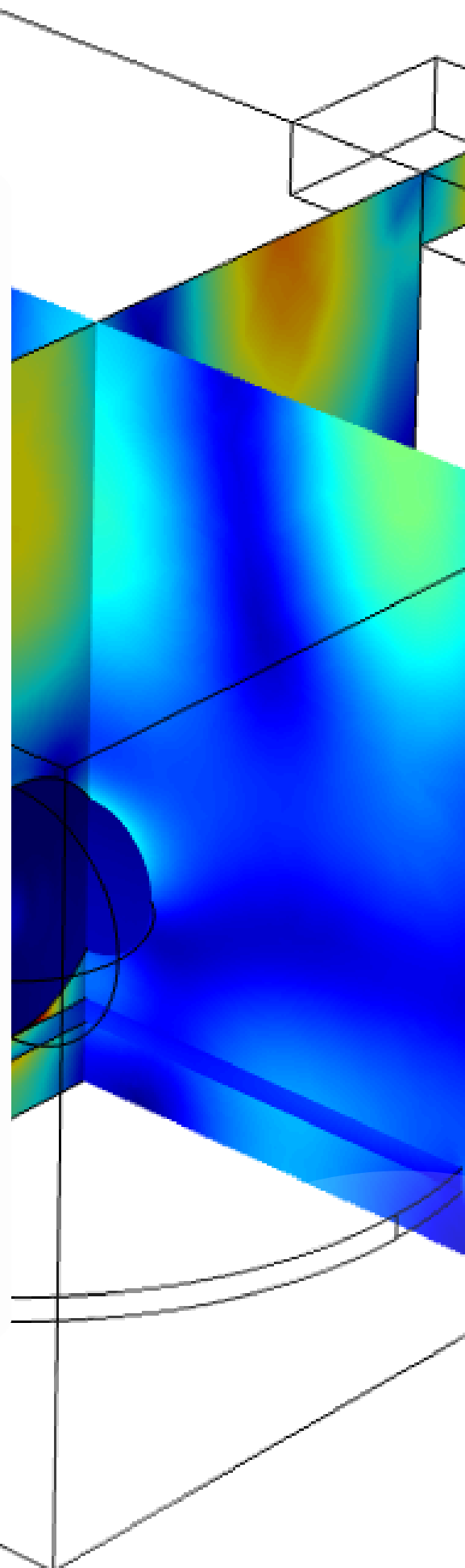
1. K-means
2. DBSCAN
3. GMM
4. DL: CNN/LSTM/SVM

Achieving >80% clustering accuracy.

## Innovation & Optimization

Develop and implement a **novel** clustering method by modifying and creating ensembles of existing models.

Demonstrate at least 90% clustering accuracy and a quantifiable improvement in terms of computational efficiency from the best performing method of Goal 2



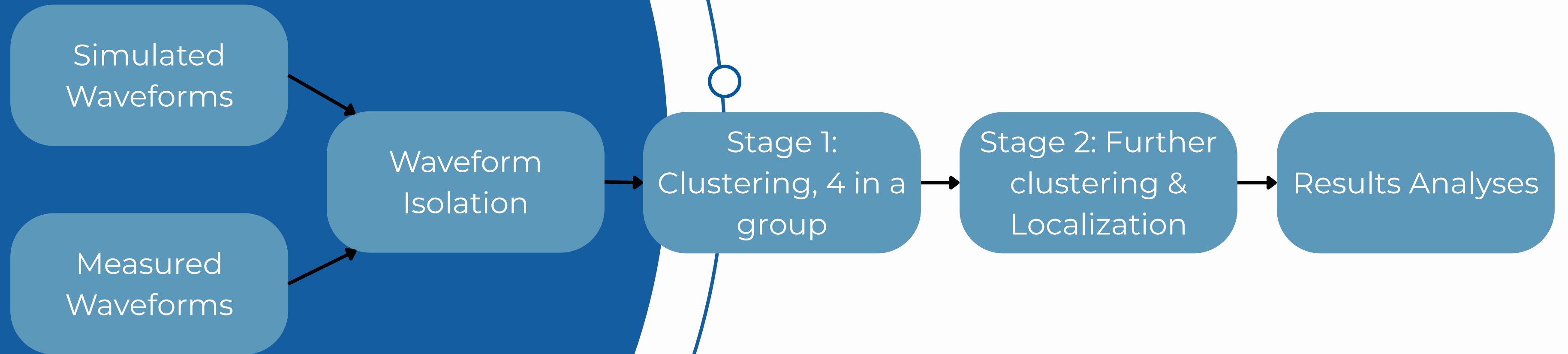
# Literature Review

| Reference                                                                                                  | Technique                                                   | Purpose                                               |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|
| He et al., 2020<br>Qin et al., 2020                                                                        | Gaussian current impulse in 3D EM wave solver               | Multiple PD separation                                |
| Loubani et al., 2019, Zanjani et al., 2012<br>Zheng et al., 2023                                           | Gaussian current impulse/DBD in 3D EM wave solver           | PD propagation characteristics                        |
| Umemoto & Tenbohlen, 2018<br>Toshihiro et al., 2008<br>Jaber et al., 2012                                  | Gaussian current impulse/preset signal in 3D EM wave solver | Comparing simulated & propagated waveforms            |
| Tian & Kawada, 2014<br>Ohtsuka et al., 2006                                                                | Gaussian/geometric current impulse in 3D EM wave solver     | Determining PD characteristics (DOA, charge quantity) |
| Len et al., 2022<br>Ma et al., 2002<br>Sahnoune et al., 2024<br>Kumar et al., 2023<br>Friebe & Jenau, 2025 | Equation based PD simulation (SEDO, DEDO, DEP, DOP)         | Denoising                                             |

| Reference                                                                       | Technique                                                                     | Results                                                                  |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Hao et al., 2011                                                                | WD + PCA + DBSCAN                                                             | Successfully obtained clusters of current waveforms from IEC60270 method |
| Pinpart & Judd, 2010                                                            | Envelope Analysis                                                             | Successfully clustered UHF PD sources with margin of 10:1                |
| Almehdhar & Prochazka, 2025                                                     | Time-frequency analysis + DL                                                  | 100% experimental accuracy                                               |
| Yang & Judd, 2003                                                               | Wavelet Analysis                                                              | Successfully clustered 2 real PD sources (UHF) with 100% accuracy        |
| Carvalho et al., 2024                                                           | Statistical features + ML (SVM, GMM, K-means, Mean Shift)                     | 90-93% Accuracy for UHF PD signals                                       |
| Fu et al., 2024                                                                 | T-F spectra + t-SNE                                                           | 100% clustering accuracy, but 10 times computational cost                |
| Z. Chen et al., 2025                                                            | Akima cubic interpolation + SELMS + time, freq, t-f features + t-SNE + DBSCAN | Successfully separated all 4 discharge types                             |
| Zhang et al., 2022, Rauscher et al., 2024, L. Chen et al., 2025, Fuet al., 2025 | Deep learning: ResNet18, FCNN, CNN, LSTM, BiLSTM, GRU, Adaptive VMD,          | >90% clustering accuracy                                                 |



# Methodology: Flowchart



# Digital Twin

SEDO

$$V_0 e^{-\frac{(t-t_d)}{\tau}} \sin(2\pi f_c(t-t_d) + \theta)$$

DEDO

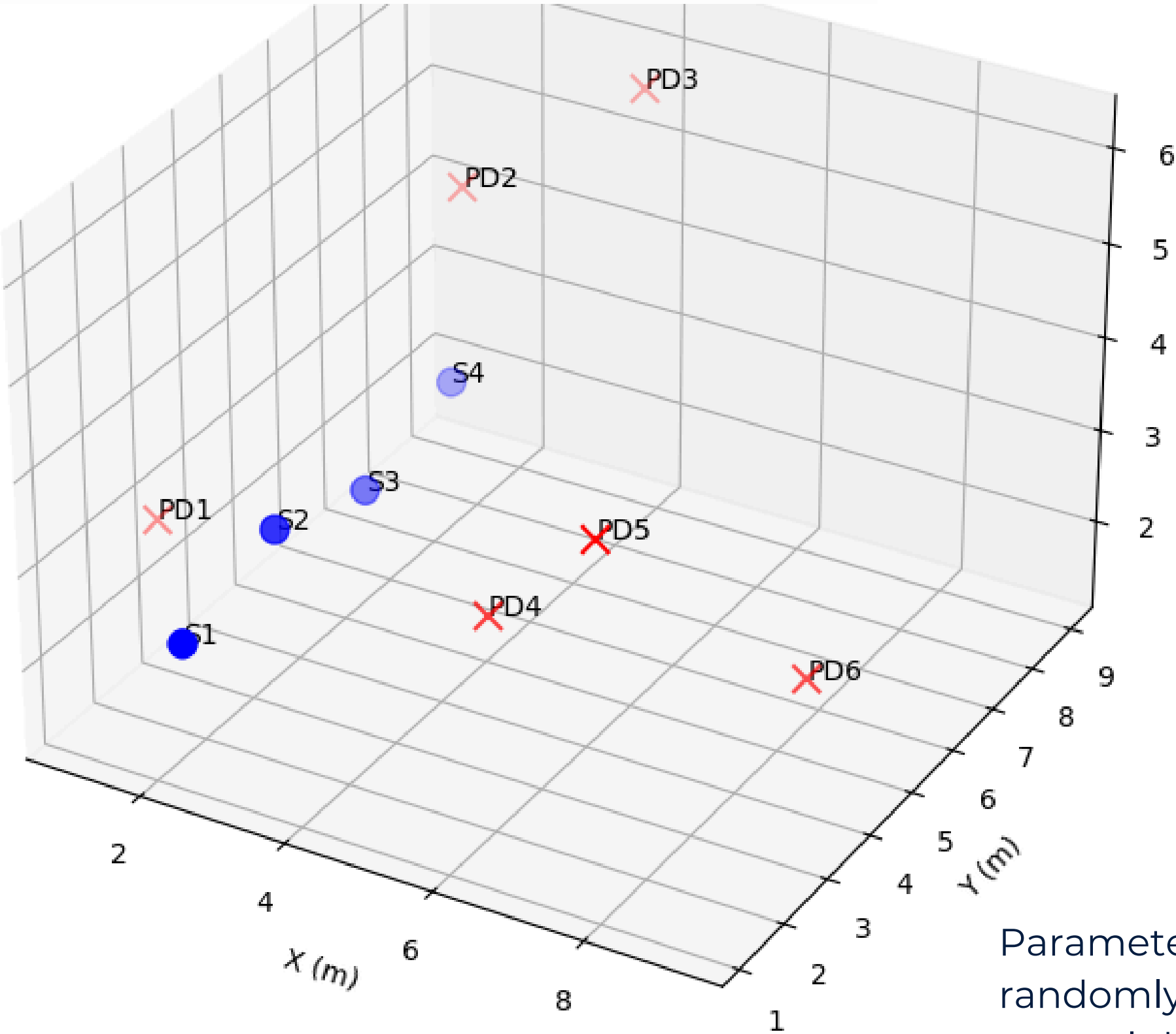
$$V_0(e^{-\alpha(t-t_d)} - e^{-\beta(t-t_d)})$$

DEP

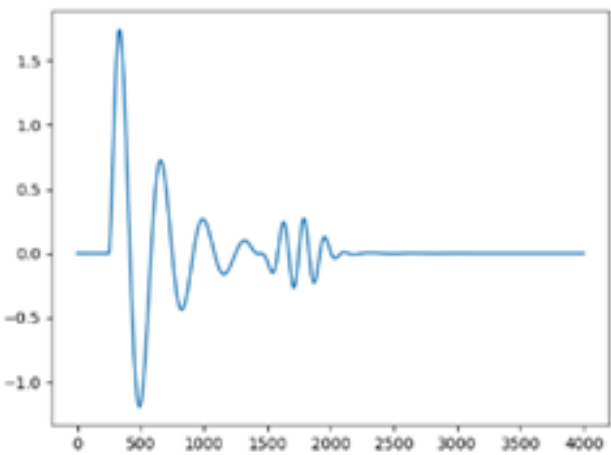
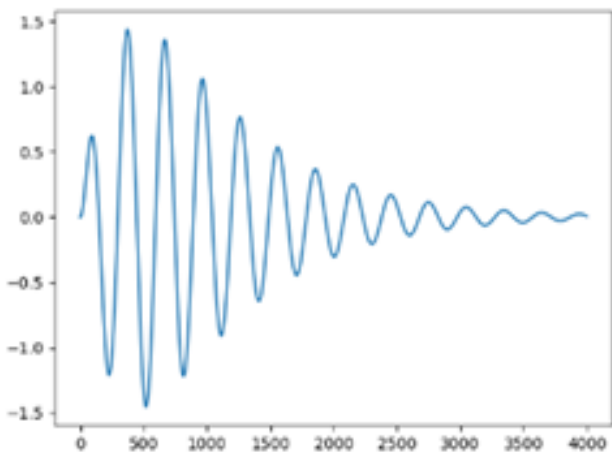
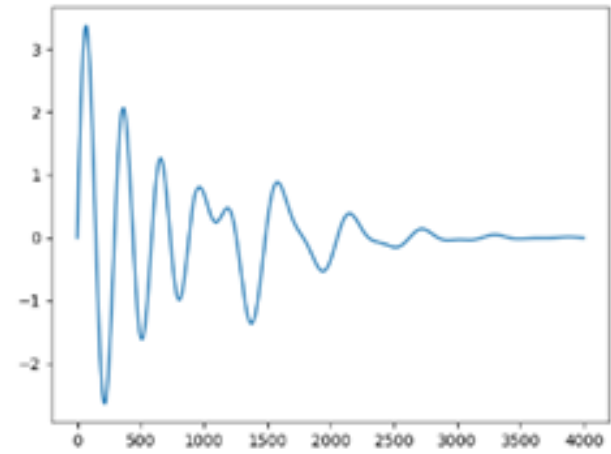
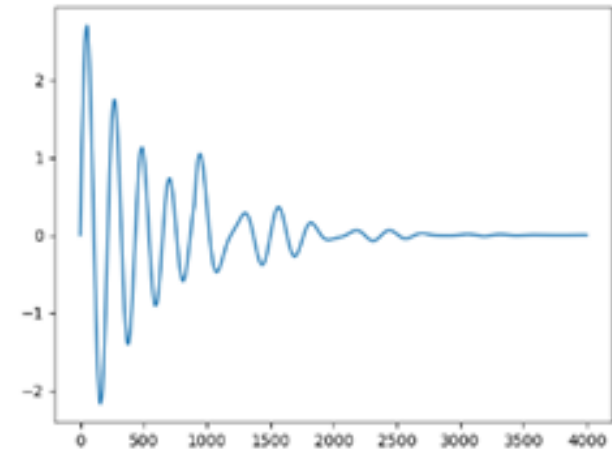
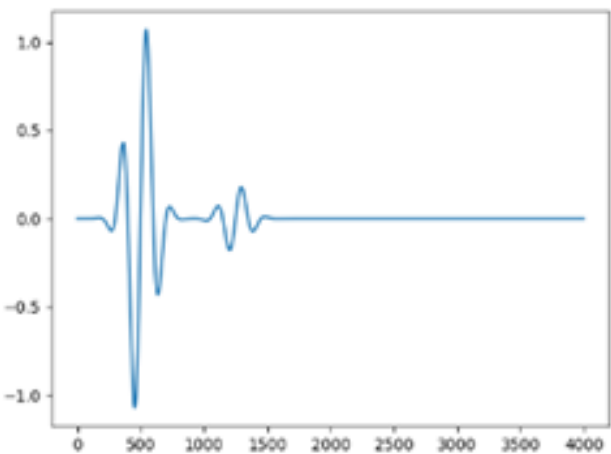
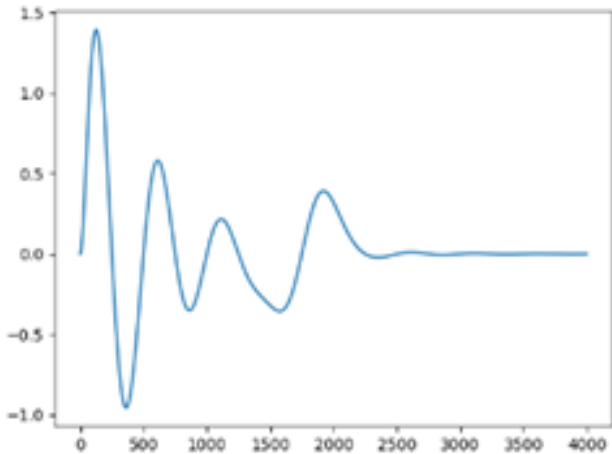
$$V_0(e^{-\alpha(t-t_d)} - e^{-\beta(t-t_d)}) \sin(2\pi f_c(t-t_d) + \theta)$$

SMG

$$Ae^{-\frac{((t-t_0-t_d)^2)}{2(\tau^2)}} \sin(2\pi f_c(t-t_0-t_d))$$

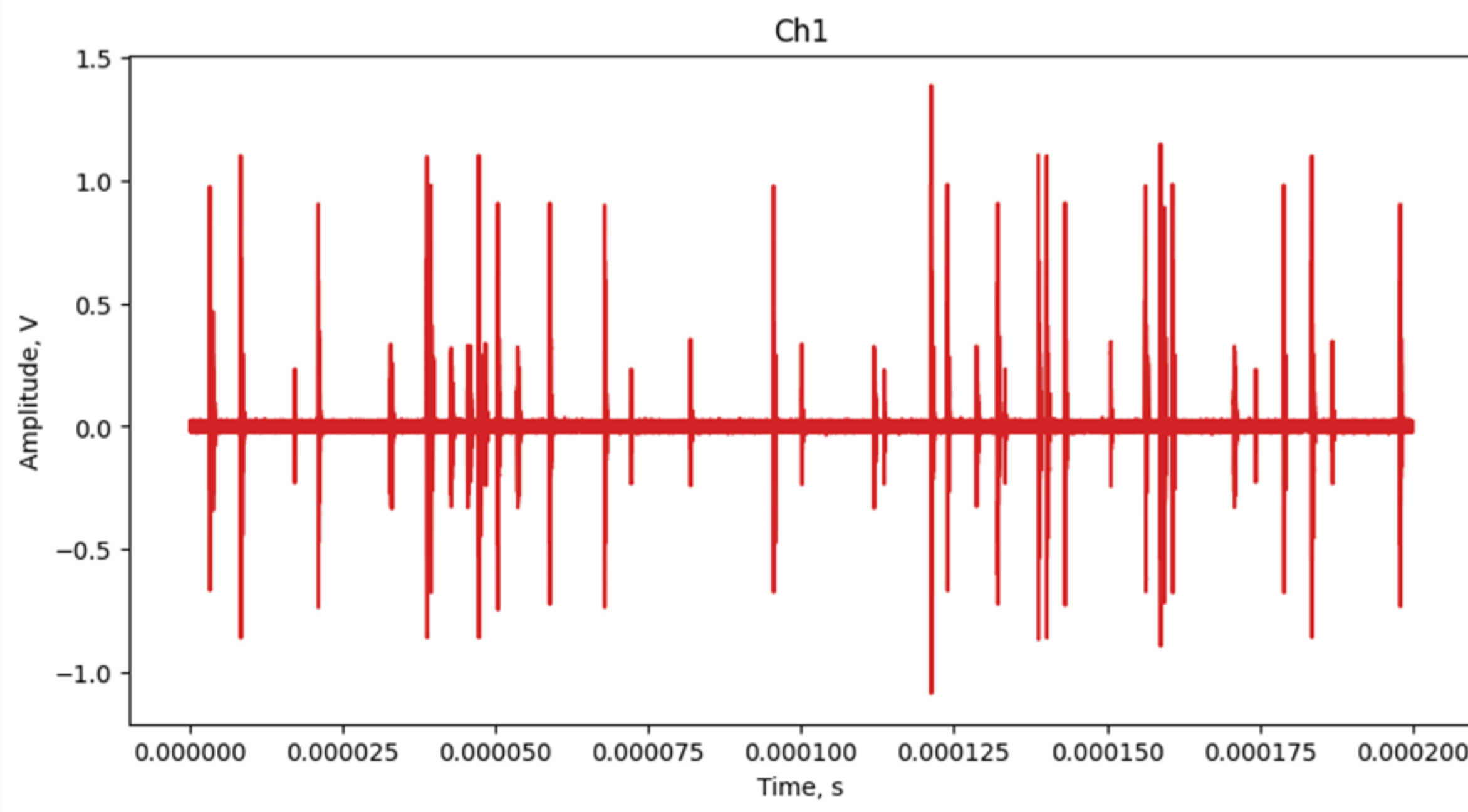


Parameters were initialized randomly, according to a normal distribution



# Digital Twin

+ Noise (SNR=20dB) & attenuation, 8 signals per PD,  
account for TDOA  
4 channels (mimic real life)



# Experimental Measurement

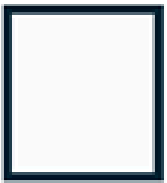
Legend:



2 PD coordinates



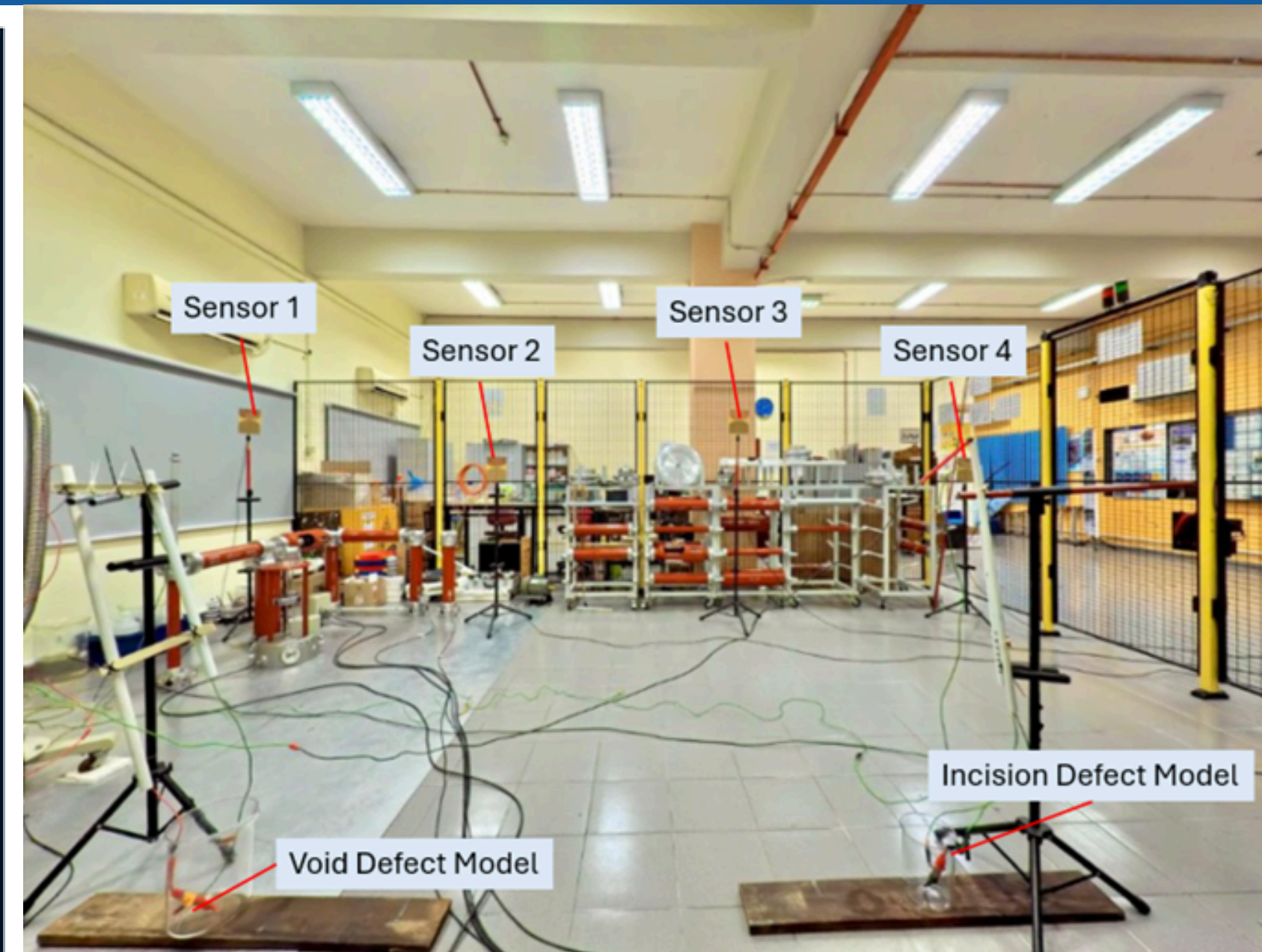
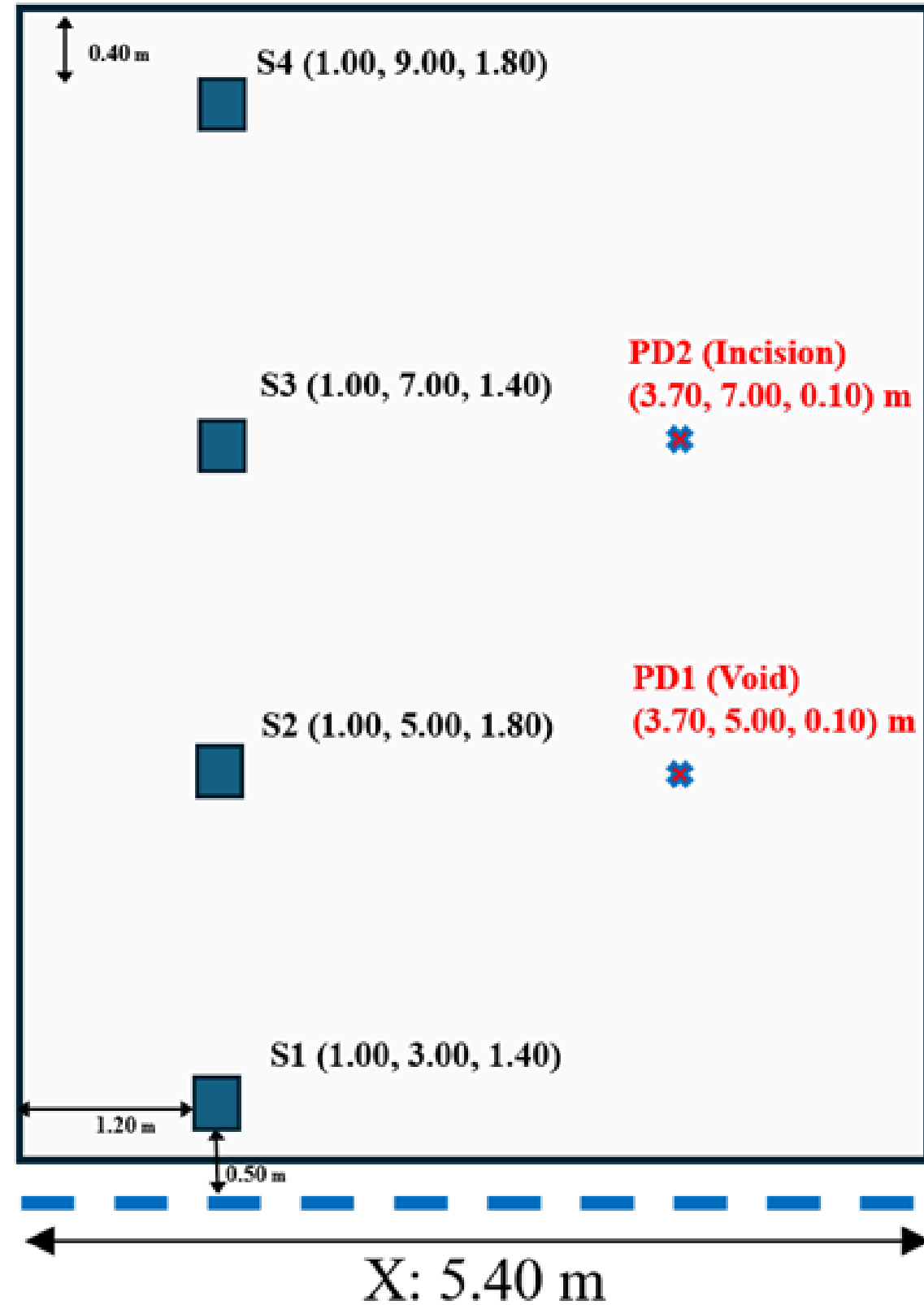
Sensors



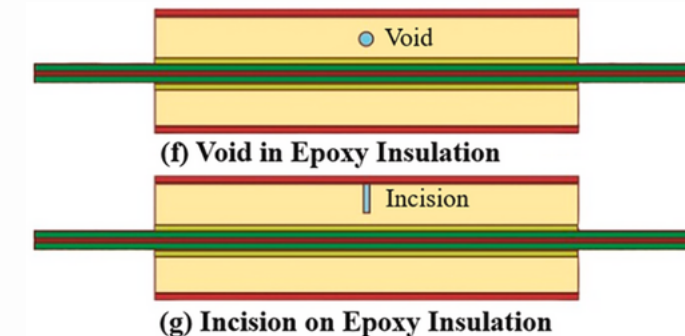
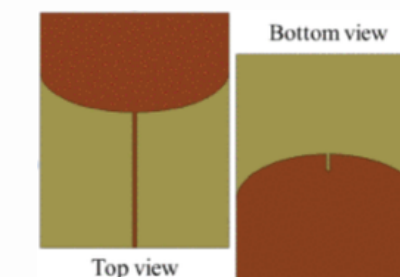
Testing area

Y: 6.87 m

Wall

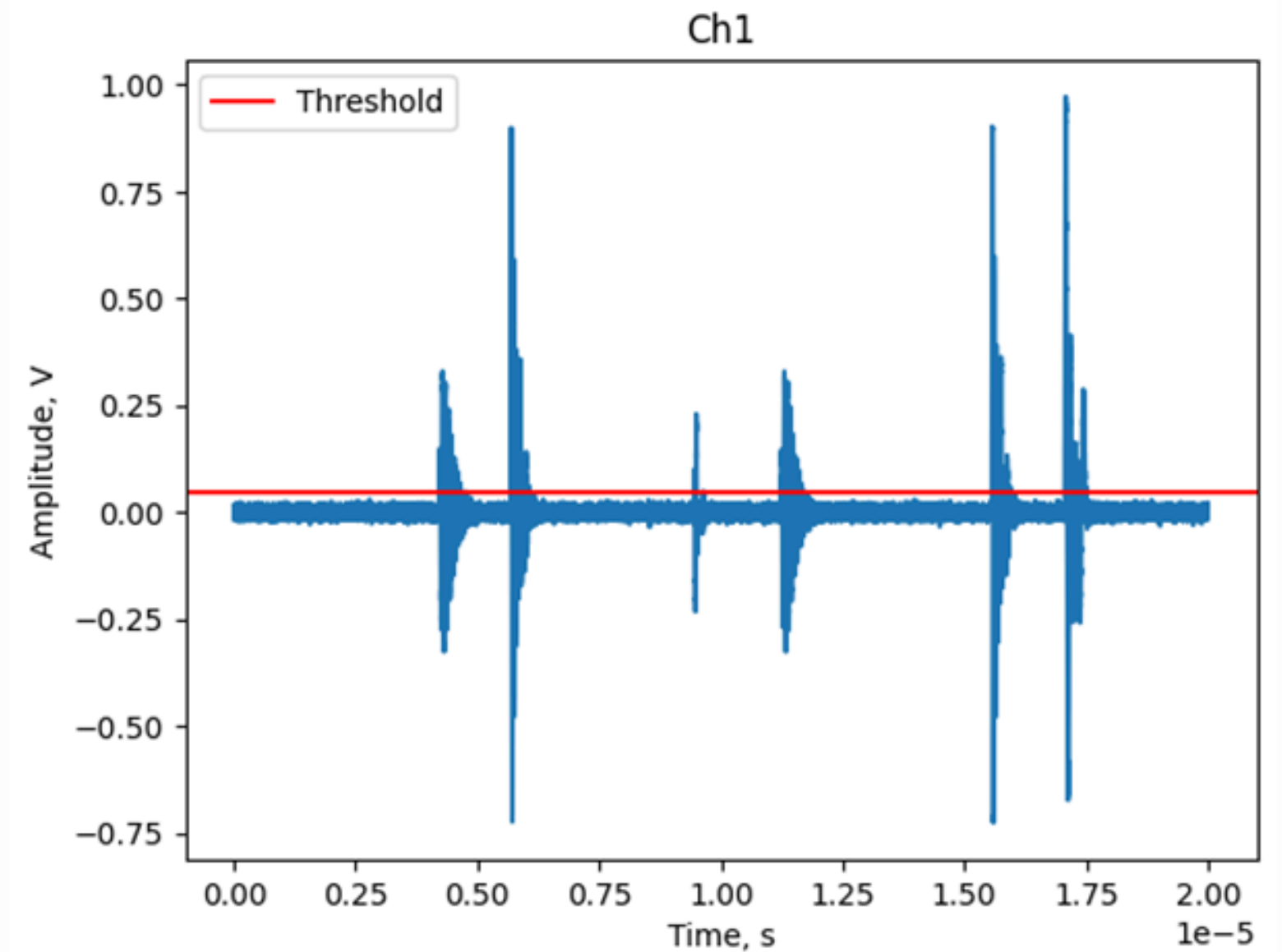
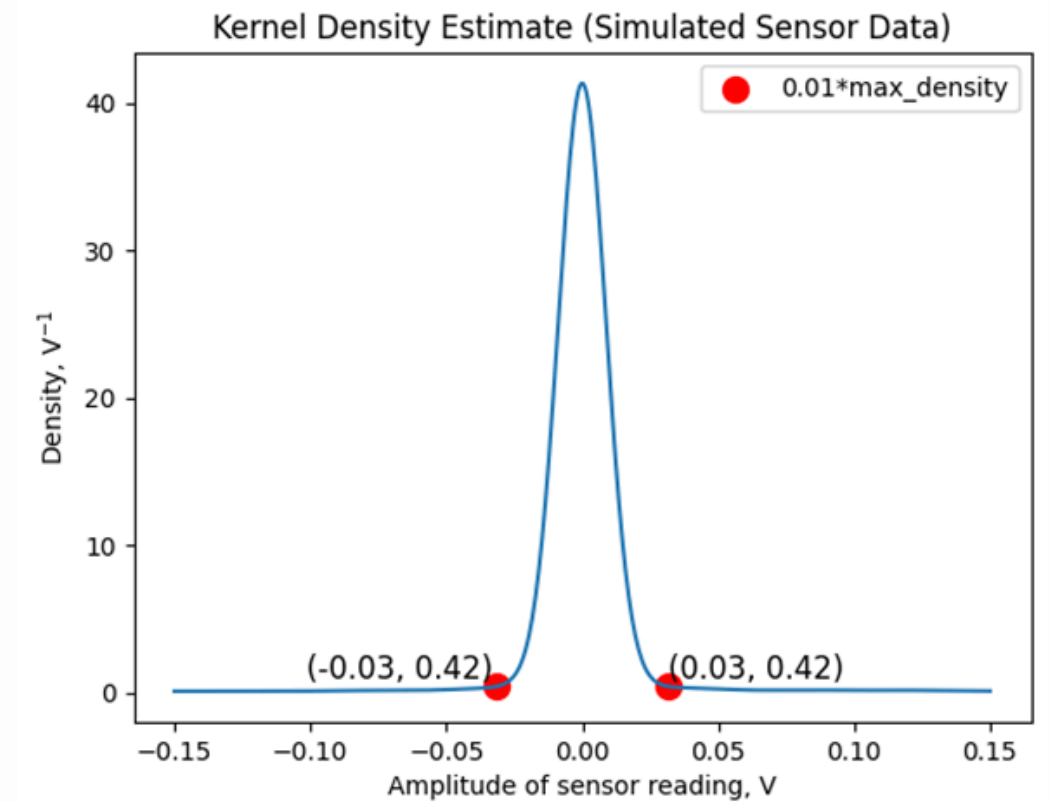
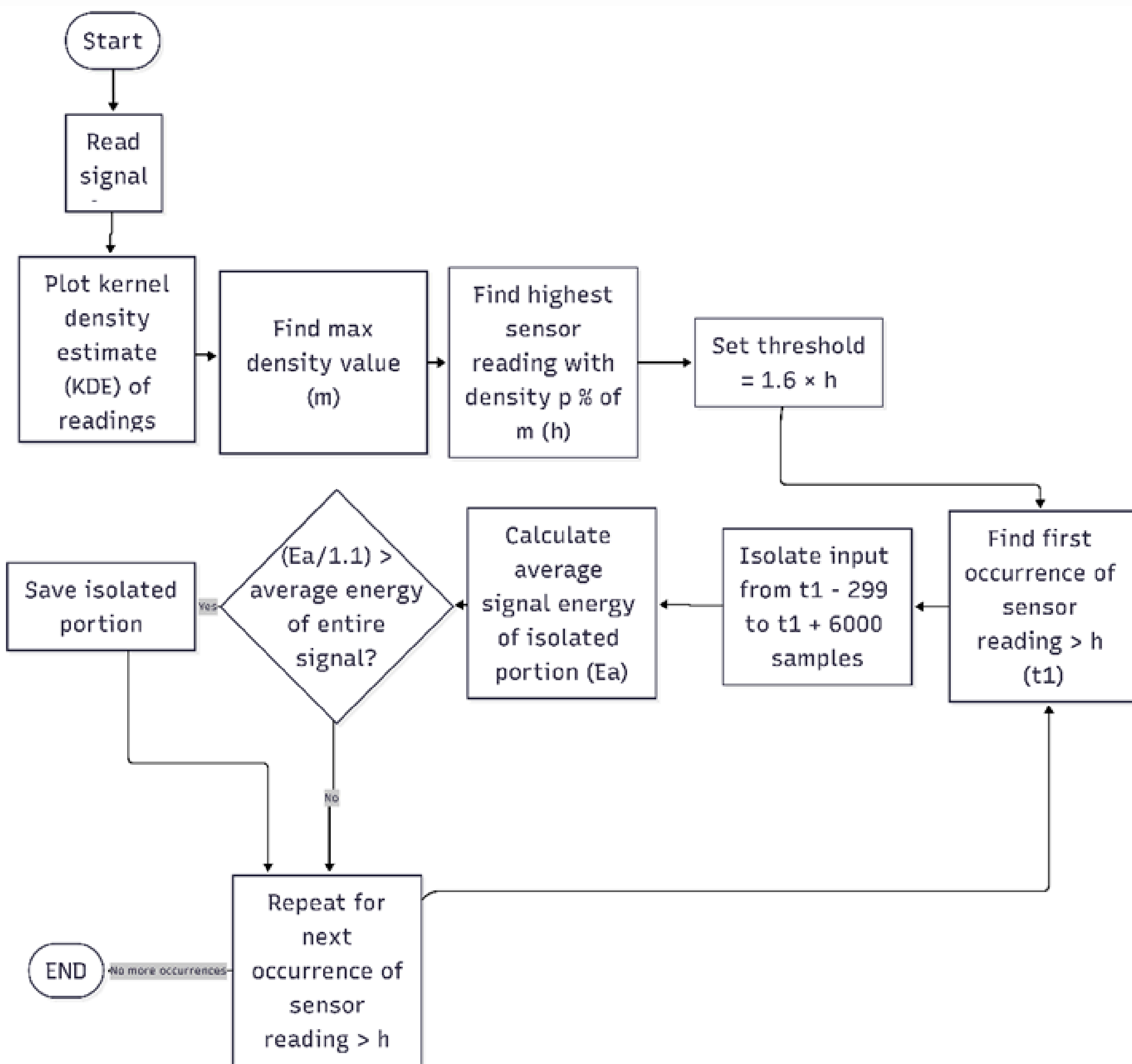


PD types: Void Defect (8kV), Incision Defect (17kV) (Madhuranga et al., 2025)  
UHF sensor: Planar monopole antenna (PMA) (Kayser Azam et al., 2024)  
10GS/s





# Waveform Isolation



# Stage 1



Stage 1 (Clustering) accuracy:  $\frac{\text{Number of correct groups}}{\text{Total number of groups formed}} \times 100\%$

## Stage 2

Further clustering & localization:

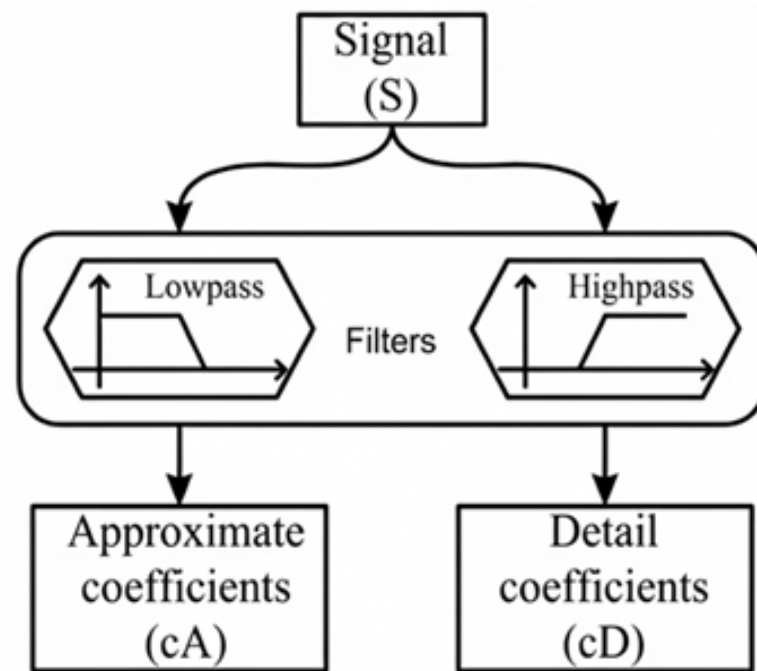
- Clustering on TDOA
- Localization on clustered TDOA
- vice-versa

$$PD \text{ Number Accuracy Ratio} = \frac{\text{Number of PDs obtained}}{\text{Actual number of PDs}}$$

$E = \text{Total euclidean distance from all estimated PD to actual PD}$

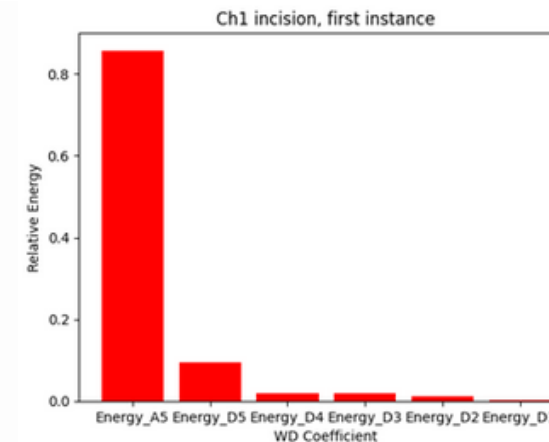
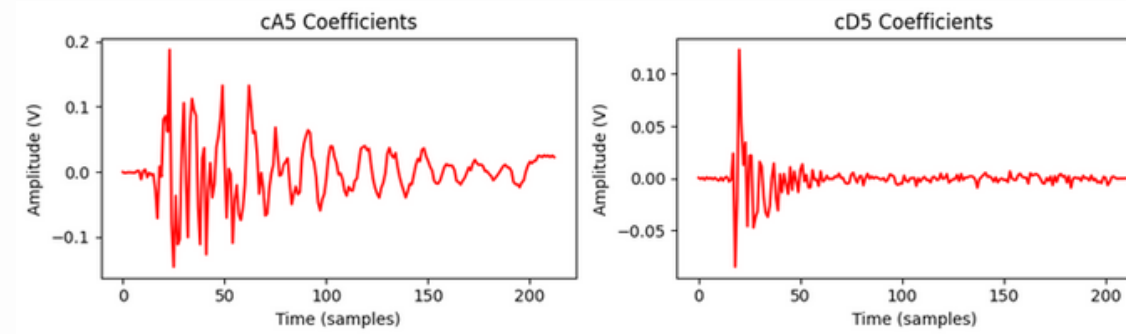
$$Avg. Euclidean Distance = \frac{E}{\text{Estimated number of PDs}}$$

# Wavelet Decomposition + PCA (Hao et al., 2011)



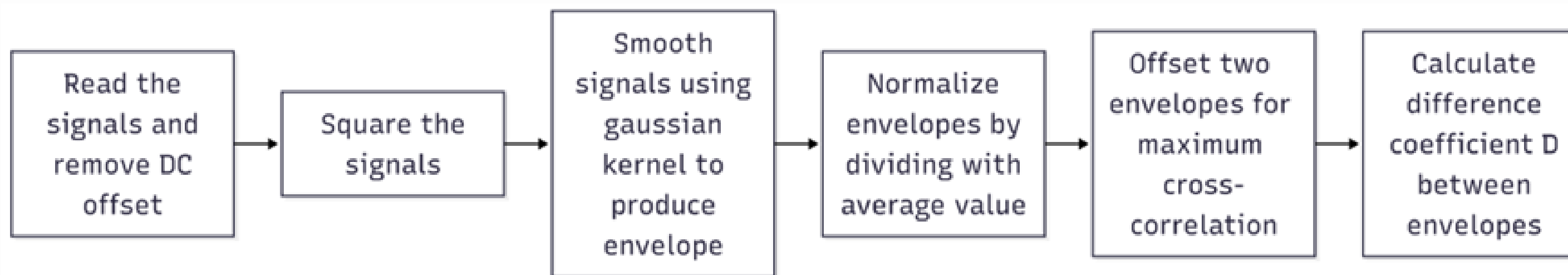
$$T_{\phi}[0,0] = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \phi(x) * f[x]$$

$$T_{\psi}[j,k] = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \psi_{j,k}(x) * f[x]$$



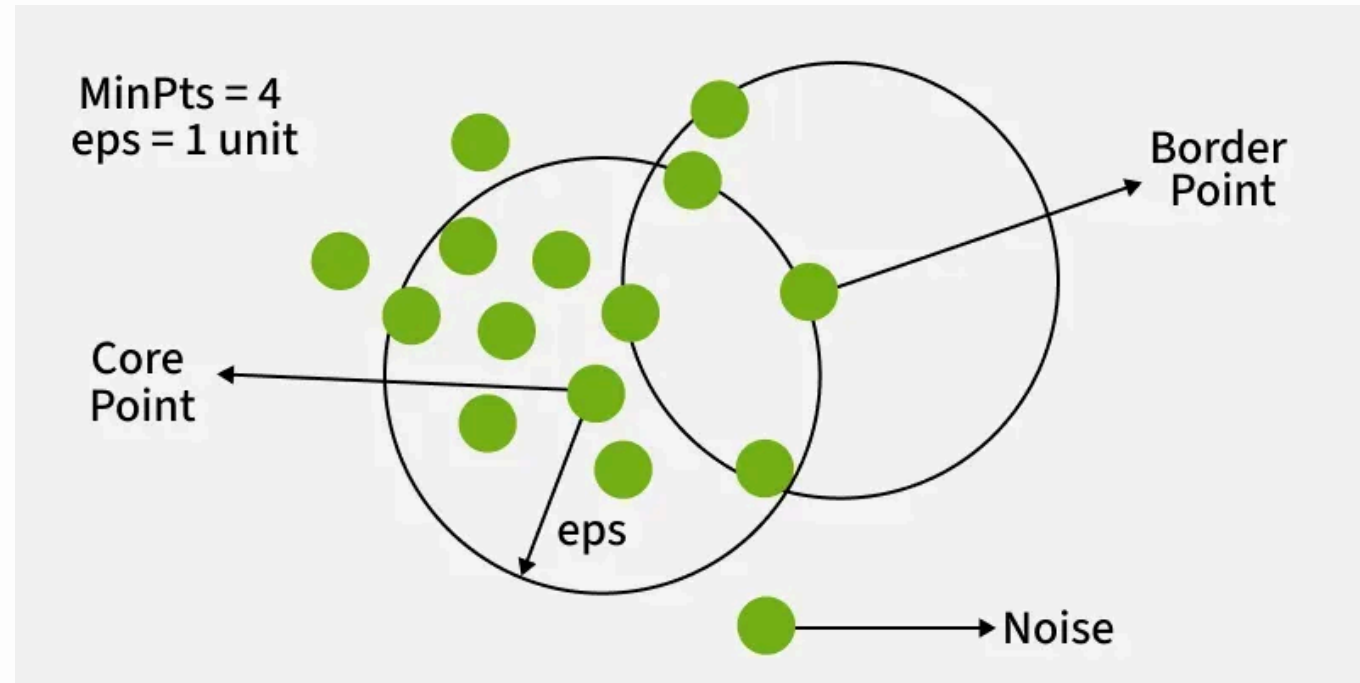
PCA → **3 Features**

# Envelope Detection (Pinpart & Judd, 2010)



$$D = \sum_{i=0}^{N-1} (\Delta E N_i)^2$$

# Density Based Spatial Clustering of Applications with Noise (Hao et al., 2011)



Source: GeeksforGeeks (2025)

## Localization: Particle Swarm Optimization (PSO)

$$V_i^{k+1} = wV_i^k + c_1r_1(P_{best,i} - X_i^k) + c_2r_2(G_{best} - X_i^k)$$

$$X_i^{k+1} = X_i^k + V_i^{k+1}$$

Obtaining the minimum of an objective function by means of moving particles within the n-dimensional space of variables to optimize

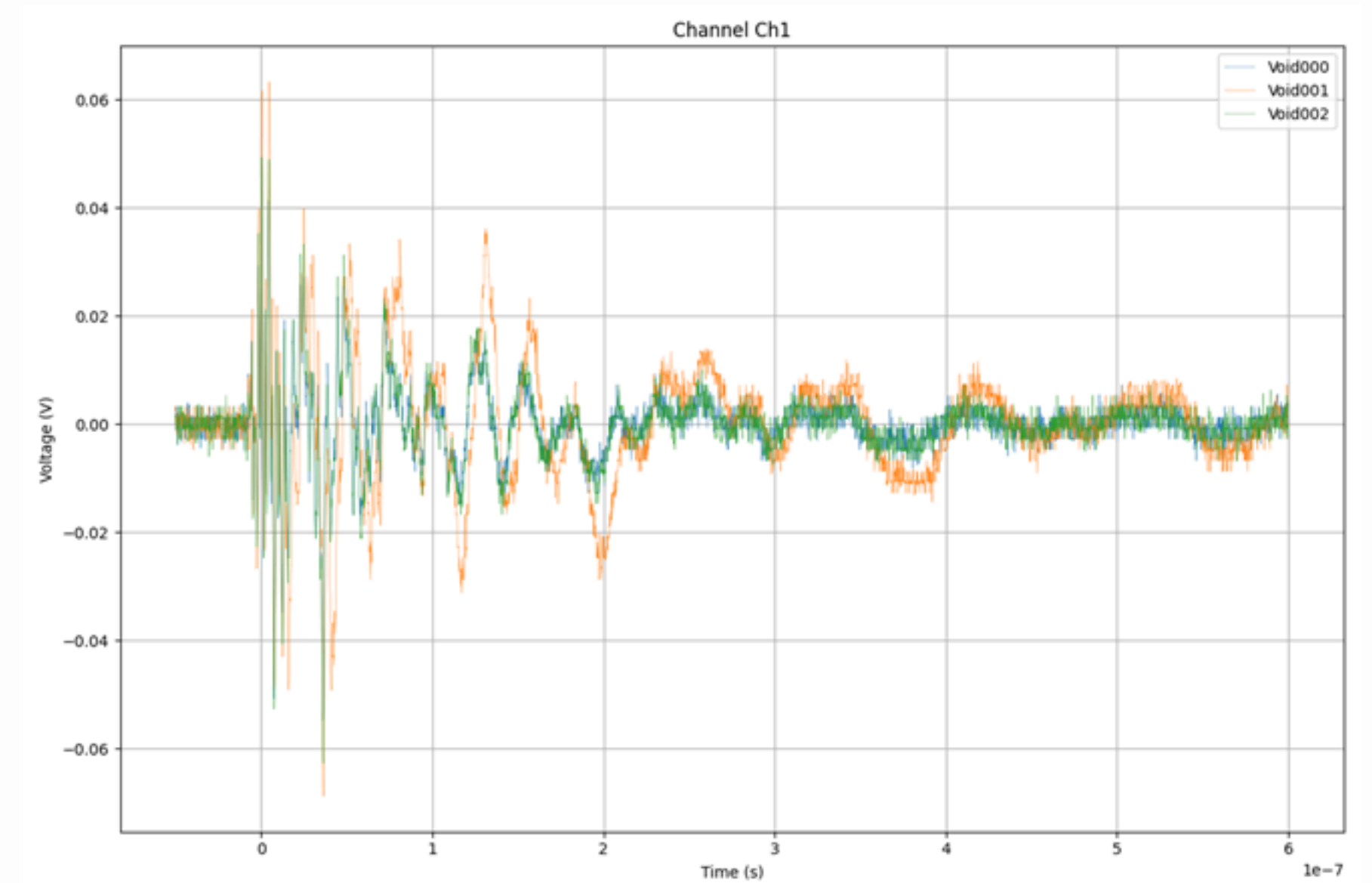
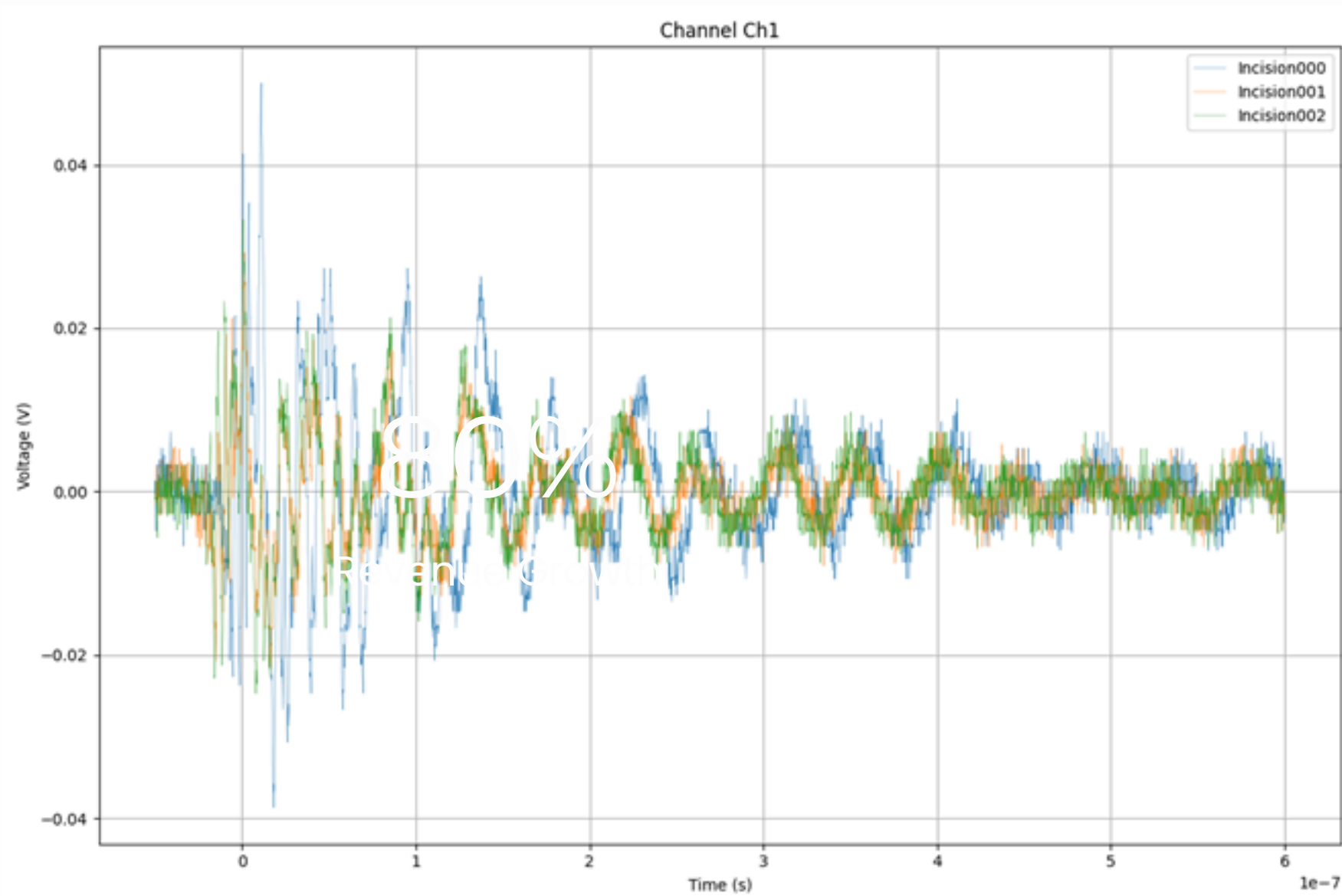




# PRELIMINARY RESULTS



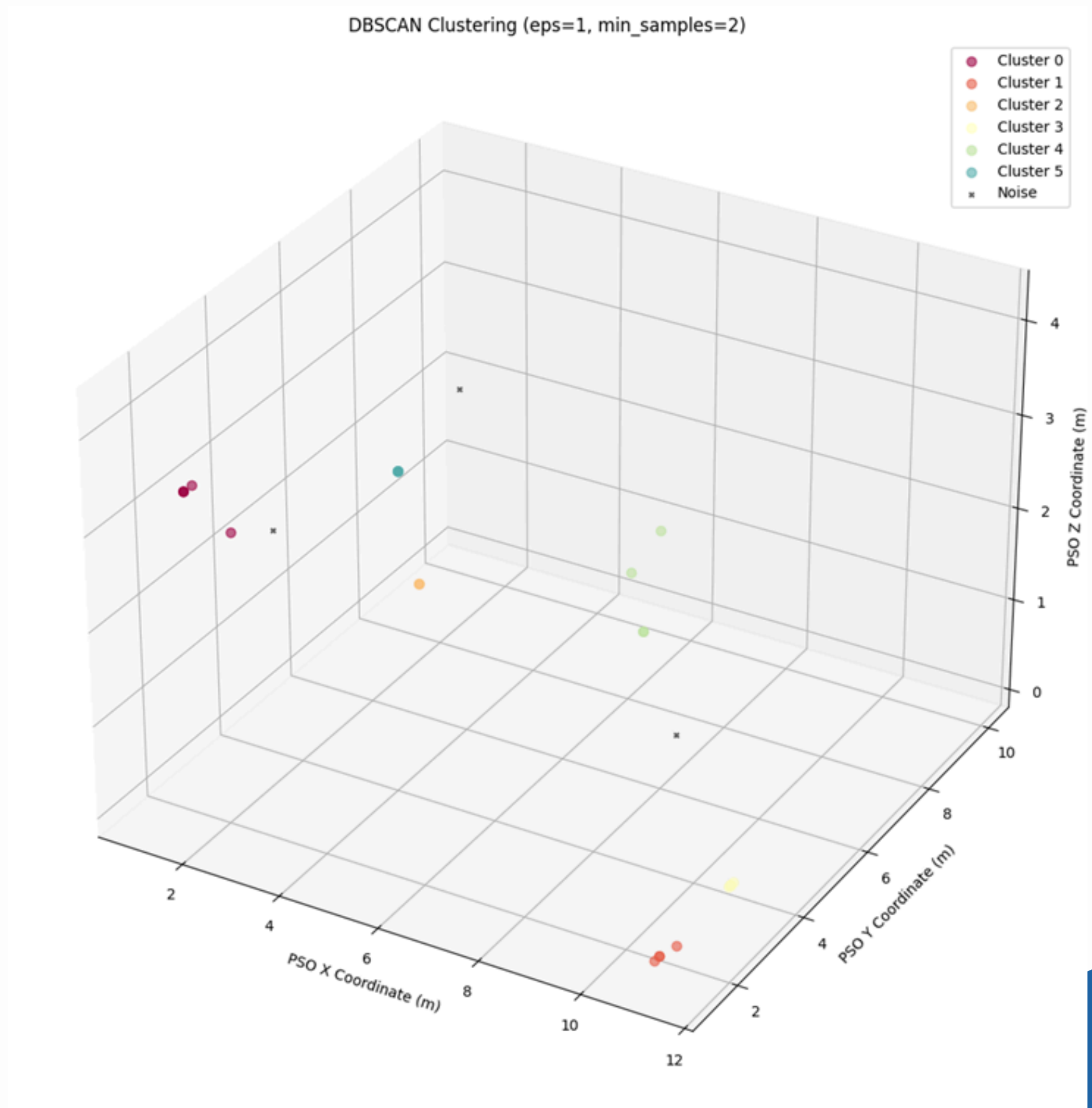
# Measured Waveforms



# Simulated: EA-> PSO-> DBSCAN

| cluster (PSO -> DBSCAN) | Euclidean Distance (m) |
|-------------------------|------------------------|
| 0                       | 0.0674388612003496     |
| 1                       | 10.8943371184299       |
| 2                       | 0.417372735094184      |
| 3                       | 9.86461479227648       |
| 4                       | 0.203562029858223      |
| 5                       | 2.11362248284787       |

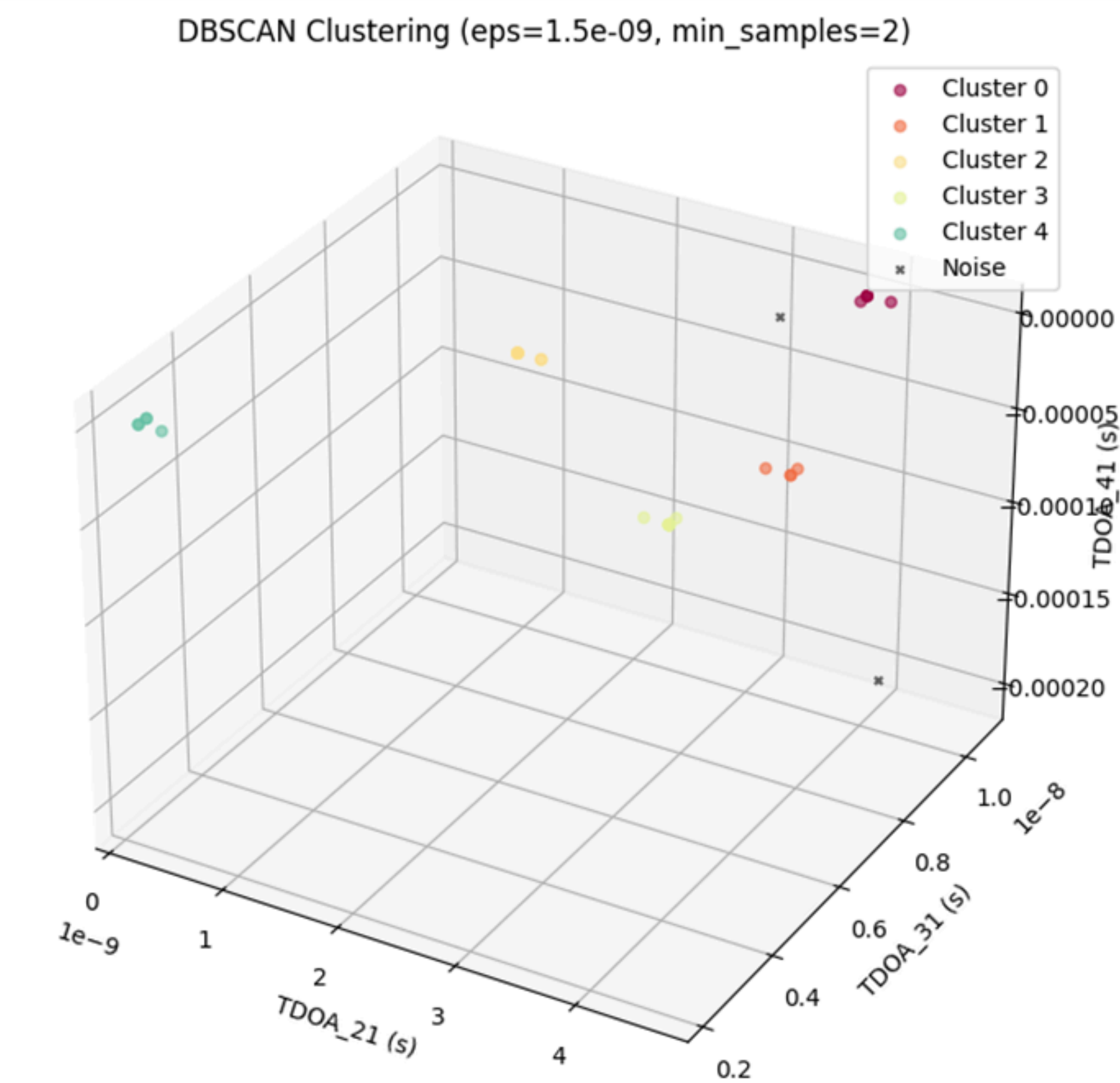
|                          |                       |
|--------------------------|-----------------------|
| Stage 1 Accuracy         | 96.77% (30 out of 31) |
| PD Number Accuracy Ratio | 6/6                   |
| Avg. Euclidean Distance  | 3.9268247 m           |



# Simulated: EA-> DBSCAN-> PSO

| cluster (DBSCAN-> PSO) | Euclidean Distance (m) |
|------------------------|------------------------|
| 0                      | 0.132664991614215      |
| 1                      | 10.8359078992025       |
| 2                      | 2.14758003343298       |
| 3                      | 5.52467193596144       |
| 4                      | 1.13617780298683       |

|                          |                       |
|--------------------------|-----------------------|
| Stage 1 Accuracy         | 96.77% (30 out of 31) |
| PD Number Accuracy Ratio | 5/6                   |
| Avg. Euclidean distance  | 3.95540 m             |





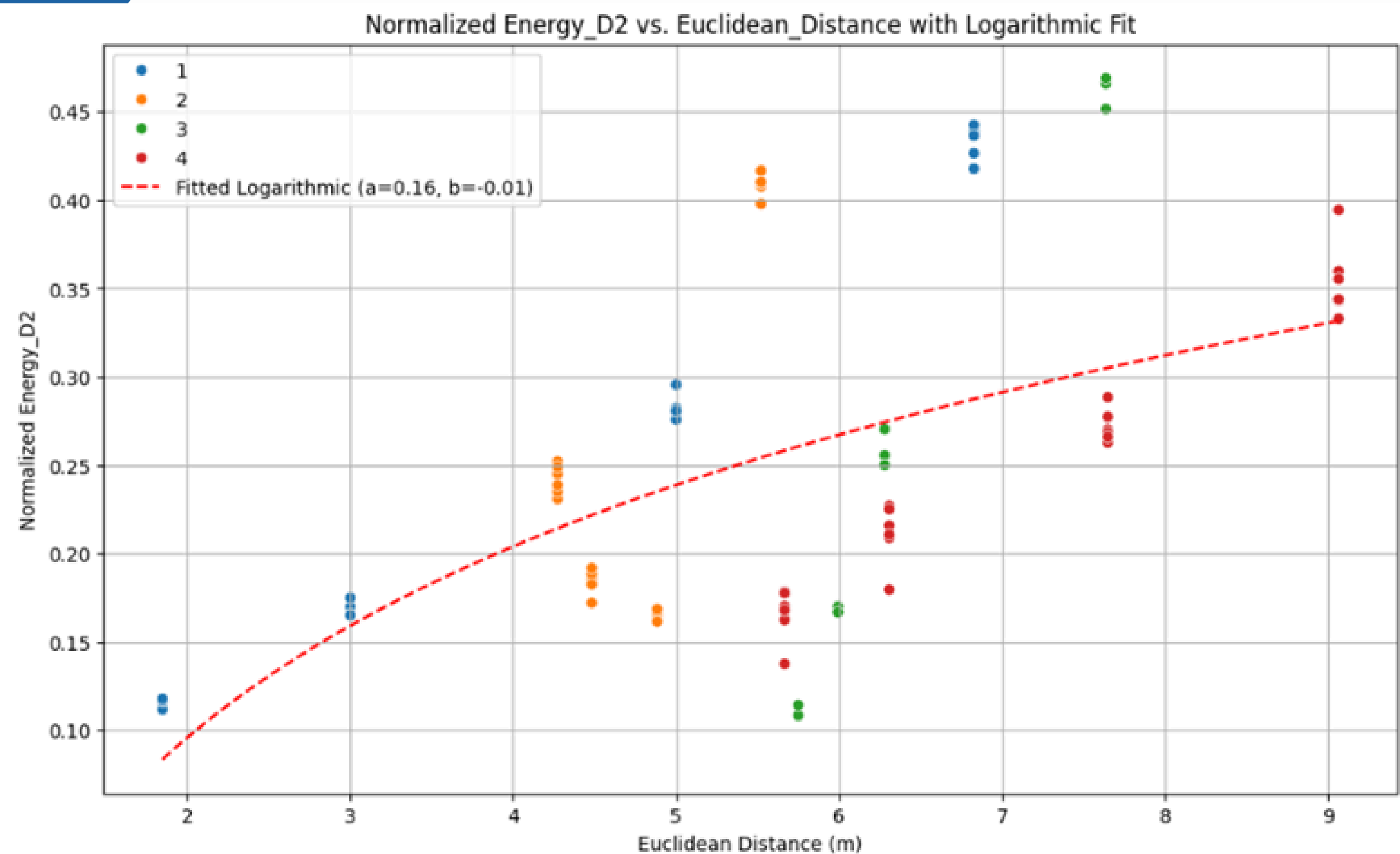
# Simulated: WD+PCA-> PSO-> DBSCAN

|                  |                      |
|------------------|----------------------|
| Stage 1 Accuracy | 12.90% (4 out of 31) |
|------------------|----------------------|

| cluster (DBSCAN-> PSO) | Est. X (m) | Est. Y (m) | Est. Z (m) | Actual X (m) | Actual Y (m) | Actual Z (m) | Euclidean Distance (m) |
|------------------------|------------|------------|------------|--------------|--------------|--------------|------------------------|
| 4                      | 11         | 3.57       | 0.1        | 9.2          | 3.5          | 2.8          | 3.24575106870505       |

# Simulated: WD+PCA-> DBSCAN-> PSO

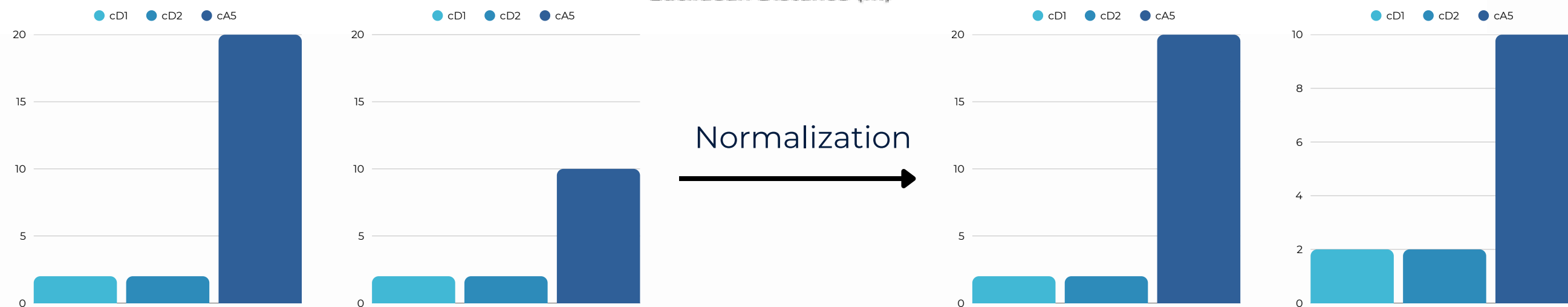
| cluster (DBSCAN-> PSO) | Est. X (m) | Est. Y (m) | Est. Z (m) | Actual X (m) | Actual Y (m) | Actual Z (m) | Euclidean Distance (m) |
|------------------------|------------|------------|------------|--------------|--------------|--------------|------------------------|
| 0                      | 11         | 3.56       | 0.21       | 9.2          | 3.5          | 2.8          | 3.15463151572414       |



Reducing WD coefficients to energies  
Normalizing to constant total energy  
affects noise normalized energies  
cD2, cD1

Signal power attenuates based on  
distance travelled, but remains >99%  
of total energy

Greatly affects normalized cD2 & cD1,  
they have the greatest variability.

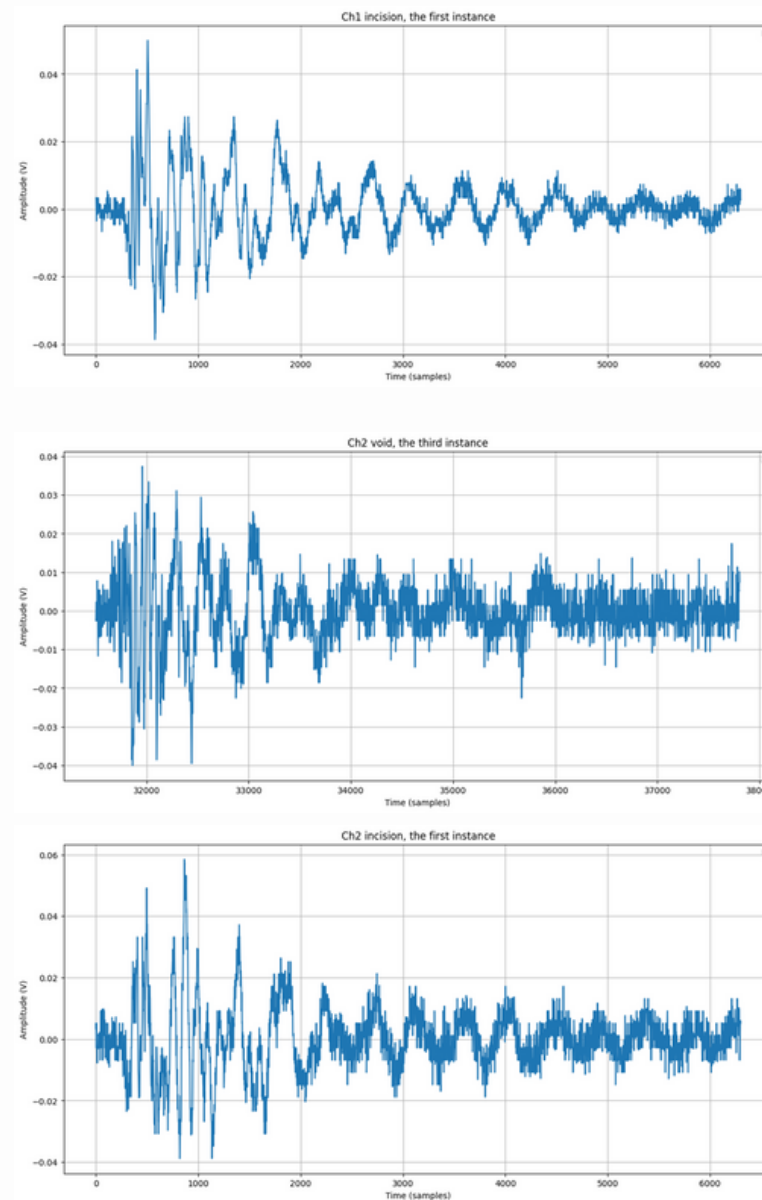


# Measured: EA/WD+PCA-> DBSCAN-> PSO/PSO-> DBSCAN

Stage 1 had 0% accuracy, for all techniques

EA failed due to the similarity of envelopes between 2 PD types, with the noise and random fluctuations in received signals causing D to be unnecessarily high

WD+PCA failed because of attenuation, and an anomaly in sensor 1 (noise power was 4x less than all other sensors). Potential flaw/situated in just the right spot for multipath propagation to cause attenuation (it was 0.5m to a wall)



| Channel | Average cD2 energy, before normalization | Average cD1 energy, before normalization |
|---------|------------------------------------------|------------------------------------------|
| 1       | 0.005528035                              | 0.000399873                              |
| 2       | 0.027646102                              | 0.001756094                              |
| 3       | 0.026863238                              | 0.001755848                              |
| 4       | 0.028222984                              | 0.001744885                              |

# Conclusion

- Current feature extraction methods do not sufficiently represent the time-domain information/shape of a signal (Difference coefficient & energy are scalars)
- Big difference between clustering accuracy of simulated & measured waveforms (>90%)
- Denoising needed to make algorithm less sensitive to noise

| Data      | Technique                           | Stage 1<br>(Clustering)<br>Accuracy | PD number<br>accuracy | Avg. Euclidean<br>Distance |
|-----------|-------------------------------------|-------------------------------------|-----------------------|----------------------------|
| Simulated | EA+PSO+DBSCAN                       | 96.77%                              | 6/6                   | 3.9268m                    |
| Simulated | EA+DBSCAN+PSO                       | 96.77%                              | 5/6                   | 3.9554m                    |
| Simulated | WD+PCA+PSO<br>+DBSCAN/DBSCAN+PSO    | 12.90%                              | N/A                   | ~3m*                       |
| Measured  | EA/WD+PCA+PSO+DBSCAN/<br>DBSCAN+PSO | 0%                                  | Failed                | Failed                     |

\*One cluster only



# Future Work

1. Better, more accurate way of modelling PD using ANSYS HFSS (He et al., 2020)
2. Implementing and comparing the rest of the clustering & feature extraction methods:
  - a. K-means
  - b. Gaussian Mixture Model
  - c. Selected bispectra
  - d. Deep Learning (CNN, LSTM, SVM)
3. Optimization, innovation by modifying and inventing new methods

**THANK YOU**